Obaidullah Miakhil

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RESEARCH INTERESTS

Multilingual AI researcher (native Pashto and Dari, fluent English and Urdu) with hands-on experience building weakly-supervised deep learning systems, LLM-based applications, and uncertainty-aware models for high-stakes decision support. Seeking to extend this foundation into Natural Language Processing and Large Language Models, with particular interest in low-resource and multilingual NLP, retrieval-augmented generation, and trustworthy/explainable AI — applying methods such as calibrated uncertainty quantification and weak supervision, developed through prior work, to make language models more reliable for real-world deployment.

EDUCATION

Ph.D. — Seeking Admission, Artificial Intelligence (NLP / LLMs)

Target: Natural Language Processing, Large Language Models, Multimodal & Trustworthy AI

M.S. Artificial Intelligence *Expected: February 2027*

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology (PAF-IAST), Haripur, Pakistan

CGPA: 4.00/4.00 (two semesters completed)

Thesis: "Weakly-Supervised Localization and Subtype Categorization of Intracranial Hemorrhage: A Two-Stage Cascade Architecture" — Supervisor: Dr. Muhammad Mohsin Khan Jadoon

Relevant Coursework: Advanced Machine Learning, Advanced Deep Learning, Advanced Digital Image Processing, Time Series Analysis (completed); final semester includes Advanced NLP and related coursework

B.S. Computer Science *2025*

PAF-IAST, Haripur, Pakistan | CGPA: 3.16/4.00

Thesis: AI-Based Prediction of Insurance Claim Denials & Management

Relevant Coursework: Introduction to Machine Learning, Introduction to Deep Learning, Introduction to Artificial Intelligence — B+ or above in both lab and theory, with a major project completed in each course

Diploma, High School *2019*

Taayef Private High School, Jalalabad, Afghanistan | Final Grade: 92.98%

RESEARCH & PROFESSIONAL EXPERIENCE

Apprentice — LLM-Based Admissions Chatbot *SPCAI, Jun–Aug 2025*

- Built an LLM-based AI chatbot to support university admissions inquiries end-to-end

Generative AI Intern *Insafdaar, Abbottabad, Jul–Sep 2024*

- Developed generative AI solutions for legal-domain challenges

Research Engineer *Sino-Park Centre of Artificial Intelligence (SPCAI), Jul 2025–present*

- Project: Weakly-Supervised ICH Localization and Subtype Classification — two-stage cascade architecture (binary gatekeeper + subtype specialist), Grad-CAM++ localization, MC Dropout uncertainty quantification

Lab Engineer *Dept. of Computer Science, PAF-IAST, Jul 2025–present*

- Taught and delivered lab content for: Object-Oriented Programming, Data Structures & Algorithms, Introduction to Computer Technology, and Artificial Intelligence Lab
- Ran the Artificial Intelligence Lab entirely in Python — guided students to implement AI algorithms from scratch, from first principles through working code
- Also taught Python programming and Object-Oriented Programming — foundational teaching/mentoring experience directly applicable to course and lab work in an NLP/LLM research group

Professional BI Trainer *KPITB, Jun–Sep 2026*

- Delivered a 144-hour Business Intelligence with Power BI workshop to 50+ participants

Teaching Assistant *PAF-IAST, Jan–Jun 2025*

- Courses: Project Management; Project Management Tools

AI Researcher (Intern) *Cube Healthcare, Michigan, USA (Remote), Jul–Aug 2024*

- Developed a predictive model for insurance claim denial (93% accuracy)

Software Application Developer (Intern) *PAF-IAST Software Cell, Jul–Aug 2023*

- Built Windows LMS features; developed CMS modules with ASP.NET

PUBLICATIONS & PAPERS

- Miakhil, O. (in preparation). Weakly-Supervised Localization and Subtype Categorization of Intracranial Hemorrhage: A Two-Stage Cascade Architecture. MS Thesis, PAF-IAST.
- Miakhil, O. (manuscript in preparation). STC-RNN: Seasonal-Trend-Coupled Recurrent Neural Network for Financial Time-Series Forecasting. Target: IEEE conference/journal.
- Miakhil, O. (2026). Two-Stage AI System for ICH Detection: Reducing Radiology TAT from 38 to 21 Minutes. Abstract submitted to RSNA 2026 Annual Meeting.

RESEARCH PROJECTS

Weakly-Supervised ICH Detection (MS Thesis) *2025–present*

- Two-stage cascade: binary gatekeeper → subtype specialist. RSNA DICOM preprocessing; HU windowing; patient-level stratified splits; EfficientNet-B2 + BiLSTM; Focal Loss; MC Dropout; Grad-CAM++ localization; BHX dataset for IoU evaluation.

STC-RNN: Financial Time-Series Forecasting *2025*

- Seasonal-Trend-Coupled RNN vs. Vanilla RNN and LSTM on KSE-100 and S&P 500. GARCH-ARMA DGP; periodogram season detection; Diebold-Mariano / BH-FDR tests.

AI-Based Insurance Claim Denial Prediction (BS Thesis) *2024*

- 93% accuracy; rule-based claim scrubber; ML-claim management system integration.

Federated Learning for Gleason Grade Classification *2024*

- EfficientNet-B0 + FedProx on PANDA dataset; Kaggle T4×2; HuggingFace checkpoints.

CONFERENCES & PRESENTATIONS

- "Two-Stage AI System for ICH Detection," abstract submitted, RSNA 2026 Annual Meeting.
- Research Methodologies Workshop, University of Salford, UK, December 2025.
- Embedding Sustainability in Research & Practices, University of Salford, UK, November 2025.
- Artificial Intelligence Conference and Expo 2024, PAF-IAST, Haripur, Pakistan, May 2024.

AWARDS

- 2021 Allama Iqbal Scholarship for Afghan Students, Higher Education Commission (HEC), Pakistan — competitive, exam-based; supported BS Computer Science at PAF-IAST
- 2023 Robo Tech 01 Competition Award Winner, PAF-IAST
- 2019 Kankor Examination — merit-based placement, Ministry of Higher Education, Afghanistan

UNIVERSITY SERVICE

- 2025–present Lab Engineer & OOP Instructor, PAF-IAST
- 2024–present AI Conference & Expo Representative, PAF-IAST
- 2023–present Student Mentor, PAF-IAST Software Cell

TRAINING & CERTIFICATES

- 2025 Google Advanced Data Analytics Specialization, Google / Coursera
- 2025 Fundamentals to Become a Machine Learning Engineer, LinkedIn Learning
- 2024 The Nuts and Bolts of Machine Learning, Google / Coursera
- 2024 Build Generative AI Productivity Skills, Microsoft / LinkedIn
- 2023 Google Project Management Professional Certification, Google / Coursera
- 2023 Fundamentals of Digital Marketing Professional Certification, Google

COMPETENCIES

Software: Python (TensorFlow, Keras, PyTorch, Scikit-learn, Flask), MATLAB, SQL, C#, C++, ASP.NET, Power BI, Microsoft Office, Moodle CMS

AI/ML: Machine Learning, Deep Learning, NLP fundamentals, Generative AI / LLM applications, Time-Series Forecasting, Federated Learning, Data Analysis, Data Visualization

Medical AI: DICOM processing, HU windowing, Grad-CAM/Grad-CAM++, RSNA datasets, weakly-supervised localization, uncertainty quantification (MC Dropout) — directly transferable to trustworthy/explainable AI research

LANGUAGES

Pashto: Native | Dari (Persian): Native | English: Fluent (C1) | Urdu: Proficient (C1)

REFERENCES

Dr. Muhammad Mohsin Khan Jadoon — MS Thesis Supervisor, PAF-IAST / SPCAI — available upon request

Dr. Sohail Khan — Associate Professor & Acting Director, SPCAI, PAF-IAST — available upon request